

Topic: Diagonalization

Question 1. Find the matrix A whose eigenvalues are 1 and 4, and whose corresponding eigenvectors are $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, respectively.

Question 2. For each of the following matrices $A \in \mathcal{M}_{n \times n}(\mathbb{R})$, determine whether A is diagonalizable over $\mathbb{R}$, and diagonalize A . That is, if A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$.

(a) $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$

Question 3. (omitted from AY2022/2023)

For each of the following linear operators T on a vector space $\mathcal{V}$, determine whether T is diagonalizable over $\mathbb{R}$. If it is, find a basis $\mathcal{A}$ for $\mathcal{V}$ such that $[T]_{\mathcal{A}}^{\mathcal{A}}$ is a diagonal matrix.

(a) $\mathcal{V} = \mathbb{R}^3$ and T is defined by $T \left(\begin{bmatrix} a \\ b \\ c \end{bmatrix} \right) = \begin{bmatrix} b \\ -a \\ 2c \end{bmatrix}$.

(b) $\mathcal{V} = \mathcal{P}_2(\mathbb{R})$ and T is defined by $T(f(x)) = f(0) + f(1)(x + x^2)$.

Question 4. For each of the following real matrices, find its real eigenvalues. Determine the algebraic and geometric multiplicities of each real eigenvalue. Determine whether the matrix is diagonalizable or not. You need not diagonalize it.

(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

(b) $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(c) $C = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$.

Question 5. Let $A \in \mathcal{M}_{n \times n}(\mathbb{R})$ have two distinct eigenvalues λ_1 and λ_2 . Prove that if $\dim(\mathcal{E}_{\lambda_1}) = n - 1$, then A is diagonalizable.

Question 6. (omitted from AY2022/2023)

Let $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be defined by $T(z, w) = (z + iw, iz + w)$, where $\mathbb{C}^2$ is the complex vector space with dimension 2. Determine whether T is diagonalizable over $\mathbb{C}$.

Question 7.

(a) Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$.

(i) Verify that the eigenvalues of A are -1 and 5 with corresponding eigenvectors $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, respectively.

(ii) Find an expression for A^n , for every positive integer n .

(b) Let $A = \begin{bmatrix} -1 & 7 & -1 \\ 0 & 1 & 0 \\ 0 & 15 & -2 \end{bmatrix}$. Compute A^{11} .

(i) Verify that the eigenvalues of A are -2 , -1 and 1 with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$, respectively.

(ii) Compute A^{11} .

Question 8. Consider *Koch snowflake* sequence:

$$0, 1, 13, 133, 1261, 11605, 105469, \dots$$

Here, the numbers can be described by a recursive formula,

$$K_{n+2} = 13K_{n+1} - 36K_n, \text{ where } K_0 = 0 \text{ and } K_1 = 1.$$

Let $u_n = \begin{bmatrix} K_{n+1} \\ K_n \end{bmatrix}$.

(a) Find a matrix A such that $u_{n+1} = Au_n$.

(b) Let $u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Prove, by induction, that $u_n = A^n u_0$ for all positive integers n .

(c) Diagonalise A and use it to prove that

$$K_n = \frac{1}{5}(9^n - 4^n).$$

Remark. If you are interested, there is a “picture” of these snowflakes available here:

<https://oeis.org/A016153/a016153.png>.